

labsheet=01 PRIYANSHU SHARMA CU240250980 SECTION(B) (47) Mastery in Data Analytics and Visualizations and Career Advancement (Technical)

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
import os
```

```
# 1. LOAD IMAGE
# =====

image_path = "/content/pexels-friededia-30649280.jpg"
image = cv2.imread(image_path)

if image is None:
    print("ERROR: Image not found!")
    print("Make sure input.jpg is in the same folder as this Python file.")
    exit()

print("Image loaded successfully!")
```

Image loaded successfully!

```
# 2. IMAGE PROPERTIES
# =====

height, width, channels = image.shape

print("\n==== IMAGE PROPERTIES =====")
print("Width      :", width)
print("Height     :", height)
print("Channels    :", channels)
print("Data Type   :", image.dtype)
print("Resolution  :", width, "x", height)
```

```
==== IMAGE PROPERTIES ====
Width      : 8192
Height     : 5464
Channels    : 3
Data Type   : uint8
Resolution  : 8192 x 5464
```

```
# 3. DISPLAY ORIGINAL IMAGE
# =====

# OpenCV uses BGR, so convert to RGB for Matplotlib
image_rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

plt.figure(figsize=(6, 5))
plt.imshow(image_rgb)
plt.title("Original Image")
plt.axis("off")
plt.show()
```

Original Image



```
# 4. SAVE IMAGE AS JPEG AND PNG
# =====
```

```
cv2.imwrite("output_image.jpg", image)
cv2.imwrite("output_image.png", image)

print("\nImage saved as:")
print("output_image.jpg")
print("output_image.png")
```

Image saved as:
output_image.jpg
output_image.png

```
# 5. COLOR SPACE CONVERSION
# =====

# Grayscale
gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

# HSV
hsv = cv2.cvtColor(image, cv2.COLOR_BGR2HSV)

# LAB
lab = cv2.cvtColor(image, cv2.COLOR_BGR2LAB)
```

```
# 6. RESIZING
# =====

resized = cv2.resize(image, (400, 300))
```

```
# 7. ROTATION
# =====

height, width = image.shape[:2]

center = (width // 2, height // 2)

rotation_matrix = cv2.getRotationMatrix2D(
    center,
    45,
    1.0
)

rotated = cv2.warpAffine(
    image,
    rotation_matrix,
    (width, height)
)
```

```
# 8. HORIZONTAL FLIP
# =====

horizontal_flip = cv2.flip(image, 1)
```

```
# 9. VERTICAL FLIP
# =====

vertical_flip = cv2.flip(image, 0)
```

```
# 10. IMAGE NEGATIVE / COMPLEMENT
# =====

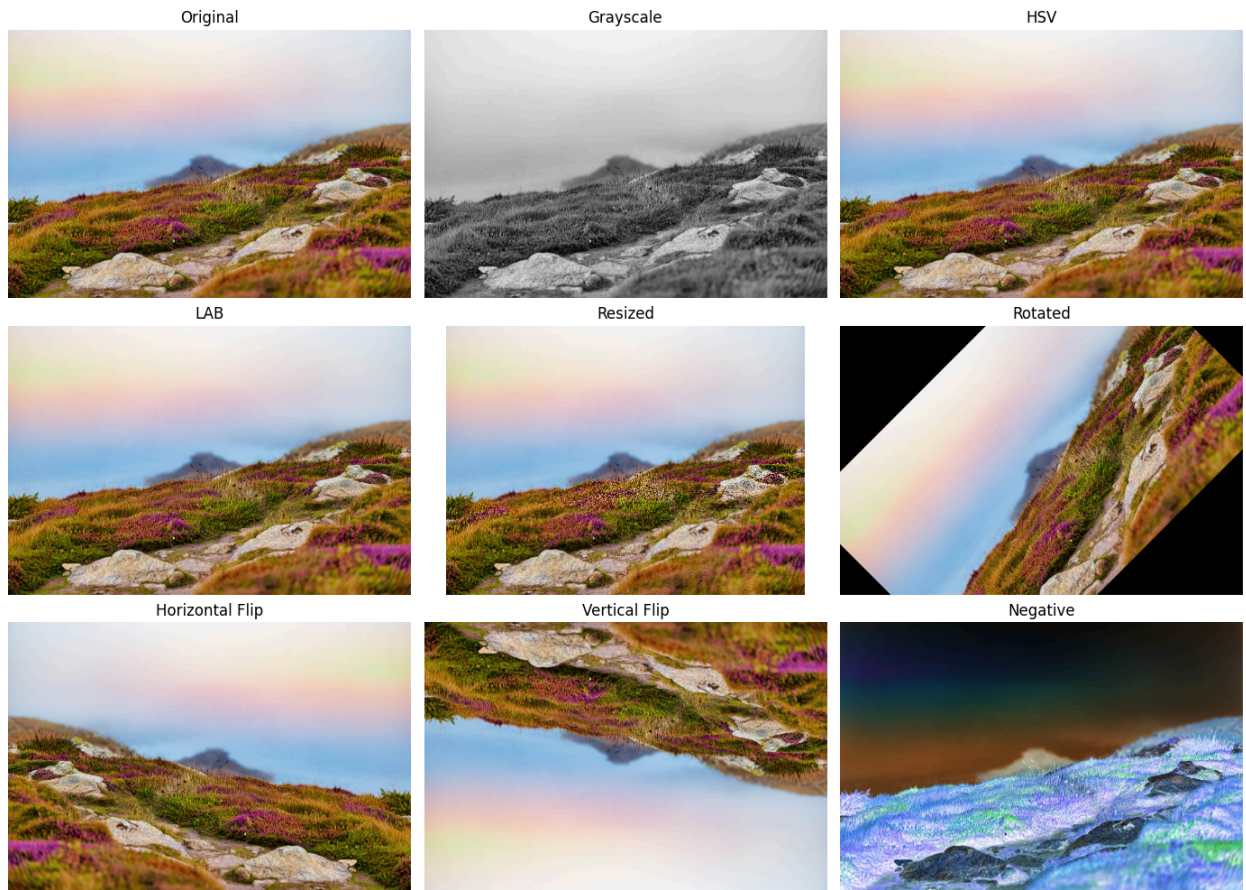
negative = 255 - image
```

```
# 11. CROP REGION OF INTEREST (ROI)
# =====

# Make sure the image is large enough
roi = image[
    height // 4 : 3 * height // 4,
```

```
width // 4 : 3 * width // 4  
]
```

```
# 12. DISPLAY ALL RESULTS  
# =====  
  
plt.figure(figsize=(14, 10))  
  
plt.subplot(3, 3, 1)  
plt.imshow(cv2.cvtColor(image, cv2.COLOR_BGR2RGB))  
plt.title("Original")  
plt.axis("off")  
  
plt.subplot(3, 3, 2)  
plt.imshow(gray, cmap="gray")  
plt.title("Grayscale")  
plt.axis("off")  
  
plt.subplot(3, 3, 3)  
plt.imshow(cv2.cvtColor(hsv, cv2.COLOR_HSV2RGB))  
plt.title("HSV")  
plt.axis("off")  
  
plt.subplot(3, 3, 4)  
plt.imshow(cv2.cvtColor(lab, cv2.COLOR_LAB2RGB))  
plt.title("LAB")  
plt.axis("off")  
  
plt.subplot(3, 3, 5)  
plt.imshow(cv2.cvtColor(resized, cv2.COLOR_BGR2RGB))  
plt.title("Resized")  
plt.axis("off")  
  
plt.subplot(3, 3, 6)  
plt.imshow(cv2.cvtColor(rotated, cv2.COLOR_BGR2RGB))  
plt.title("Rotated")  
plt.axis("off")  
  
plt.subplot(3, 3, 7)  
plt.imshow(cv2.cvtColor(horizontal_flip, cv2.COLOR_BGR2RGB))  
plt.title("Horizontal Flip")  
plt.axis("off")  
  
plt.subplot(3, 3, 8)  
plt.imshow(cv2.cvtColor(vertical_flip, cv2.COLOR_BGR2RGB))  
plt.title("Vertical Flip")  
plt.axis("off")  
  
plt.subplot(3, 3, 9)  
plt.imshow(cv2.cvtColor(negative, cv2.COLOR_BGR2RGB))  
plt.title("Negative")  
plt.axis("off")  
  
plt.tight_layout()  
plt.show()
```



```
# 13. DISPLAY ROI
# =====

plt.figure(figsize=(6, 5))
plt.imshow(cv2.cvtColor(roi, cv2.COLOR_BGR2RGB))
plt.title("Region of Interest (ROI)")
plt.axis("off")
plt.show()
```



```
# 14. SAVE PROCESSED IMAGES
# =====

cv2.imwrite("grayscale.jpg", gray)
cv2.imwrite("resized.jpg", resized)
cv2.imwrite("rotated.jpg", rotated)
cv2.imwrite("horizontal_flip.jpg", horizontal_flip)
cv2.imwrite("vertical_flip.jpg", vertical_flip)
cv2.imwrite("negative.jpg", negative)
cv2.imwrite("roi.jpg", roi)
```